

Casein Kinase II Is a Negative Regulator of c-Jun DNA Binding and AP-1 Activity

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Summary

c-Jun, a major component of the inducible transcription factor AP-1, is a phosphoprotein. In nonstimulated fibroblasts and epithelial cells, c-Jun is phosphorylated on a cluster of two to three sites abutting its DNA-binding domain. Phosphorylation of these sites inhibits DNA binding, and their dephosphorylation correlates with increased AP-1 activity. We show that two of these sites, Thr-231 and Ser-249, are phosphorylated by casein kinase II (CKII). Substitution of the third site, Ser-243, by Phe interferes with phosphorylation of the inhibitory sites in vivo and by purified CKII in vitro. Microinjection into living cells of synthetic peptides that are specific competitive substrates or inhibitors of CKII results in induction of AP-1 activity and c-Jun expression. Microinjection of CKII suppresses induction of AP-1 by either phorbol ester or an inhibitory peptide. These results suggest that one of the roles of CKII, a major nuclear protein kinase with no known functions, is to attenuate AP-1 activity through phosphorylation of c-Jun.

Introduction

The product of the *c-jun* proto-oncogene, c-Jun, is a major component of the sequence-specific transcription factor AP-1 (Bohmann et al., 1987; Angel et al., 1988a; Vogt and Bos 1990; Karin, 1991). AP-1 activity is positively regulated in response to cell stimulation with phorbol ester tumor promoters and growth factors (Angel et al., 1987; reviewed by Angel and Karin, 1991). It is also elevated in response to expression of transforming oncogenes such as v-src and Ha-ras (Imler et al., 1988; Schönthal et al., 1988; Herrlich and Ponta, 1989). c-Jun binds DNA either as a homodimer or as a heterodimer with other Jun or Fos proteins (reviewed by Vogt and Bos, 1990; Karin, 1991). While *c-jun* transcription is rapidly induced in response to the stimuli listed above, much of this regulation involves post-translational modification of preexisting c-Jun protein, which binds to a TPA response element (TRE) within the

c-jun promoter (Angel et al., 1988b; Angel and Karin, 1991). In nonstimulated fibroblasts and epithelial cells, c-Jun is phosphorylated on two to three sites next to its DNA-binding domain (Boyle et al., 1991a; Binetruy et al., 1991; Smeal et al., 1991). Phosphorylation of these C-terminal sites negatively regulates binding of c-Jun to the TRE (Boyle et al., 1991a). Treatment with phorbol esters or expression of transforming oncogenes leads to dephosphorylation of at least one of these sites (Boyle et al., 1991a; Binetruy et al., 1991). In addition, expression of transforming oncogenes stimulates the phosphorylation of c-Jun on two N-terminal sites, located within its transcriptional activation domain (Binetruy et al., 1991; Smeal et al., 1991, 1992). Phosphorylation of these sites augments the transactivating potential of c-Jun (Pulverer et al., 1991; Smeal et al., 1991) without affecting its DNA binding activity (Smeal et al., 1992).

One of the possible negative regulatory C-terminal sites, Ser-243, is substituted by Phe in v-Jun, the product of the retroviral oncogene *v-jun* (Maki et al., 1987). Substitution of Ser-243 of c-Jun by Phe increases its in vivo activity by at least 10-fold (Boyle et al., 1991a). This increase is probably due to the inhibitory effect of Phe-243 on phosphorylation of the surrounding sites, resulting in a c-Jun variant that, like v-Jun, is no longer phosphorylated at its C-terminus (Boyle et al., 1991a; Smeal et al., 1991). In addition to Ser-243, the C-terminal phosphorylation sites include Ser-249 and one of two adjacent Thr residues, 231 or 239 (Boyle et al., 1991a). In vitro, three of these residues (Thr-239, Ser-243, and Ser-249) are phosphorylated by glycogen synthase kinase 3 (GSK3), resulting in inhibition of DNA binding. However, GSK3 appears to phosphorylate c-Jun(Phe-243) or v-Jun as efficiently as wild-type c-Jun (Boyle et al., 1991a). In addition, GSK3 does not seem to be present in the nucleus (J. R. Woodgett, unpublished data, cited in Boyle et al., 1991a), whereas c-Jun is nuclear protein (Angel et al., 1989). Therefore, GSK3 is unlikely to be the major protein kinase that phosphorylates the C-terminal sites of c-Jun in vivo. Because c-Jun is a major mediator of *ras* action (Lloyd et al., 1991) and is an important nuclear target for an oncogenic phosphorylation cascade (Binetruy et al., 1991; Smeal et al., 1991, 1992), understanding its regulation by phosphorylation is essential for elucidating the biochemical pathways that transduce growth regulatory signals from the cell membrane to the nucleus. Here we present biochemical and physiological evidence that casein kinase II (CKII) is responsible for phosphorylating two of the C-terminal sites of c-Jun and is a negative regulator of its activity. The Phe-243 mutation interferes with phosphorylation of c-Jun by CKII and protects it from its inhibitory effect.

CKII is a well-characterized tetrameric Ser/Thr-specific protein kinase composed of two molecules of an α or α' catalytic subunit (relative molecular weight 40,000–45,000) and two molecules of a β -regulatory subunit (relative molecular weight 26,000) (Litchfield et al., 1990b; Meisner and Czech, 1991). Recent evidence indicates that

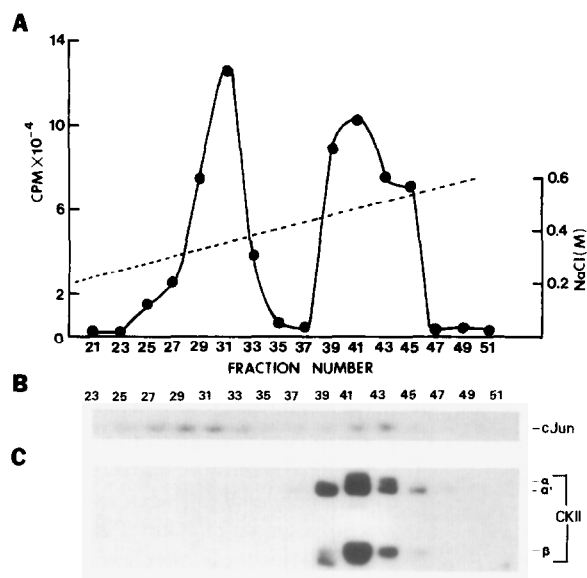


Figure 1. Fractionation of Nuclear c-Jun Kinases
Nuclear extract (5 mg of protein) of MT-v-Sis-transformed NIH 3T3 cells was chromatographed on a Mono Q column. Fractions of 0.5 ml were collected, and 20 μ l samples were assayed for c-Jun kinase activity as described in Experimental Procedures, using either a synthetic c-Jun C-terminal peptide (TPPLSPIDMESQER) (A) or purified recombinant c-Jun protein as substrates (B). (C) The indicated fractions were also immunoblotted with an anti-CKII antiserum that detects the α , α' , and β subunits of CKII.

CKII is a predominantly nuclear protein (Krek et al., 1992). Although *in vitro* CKII was shown to phosphorylate a variety of nuclear factors, including c-ErbA (Glineur et al., 1989), the E1A protein of adenovirus (Carroll et al., 1988), the E7 protein of human papilloma virus 16 (Fitzlaff et al., 1989; Barbosa et al., 1990), c-Myb (Luscher et al., 1990), the cAMP response element (CRE)-binding protein (Lee et al., 1990), p53 (Meek et al., 1990), and the serum response factor (SRF) (Manak et al., 1990), there has been little direct evidence that it regulates the activity of any transcription factor *in vivo*. While the physiological role of CKII is unknown, disruption of both the α and α' subunit genes of *Saccharomyces cerevisiae* was shown to result in lethality (Padmanabha et al., 1990). Therefore, CKII activity is absolutely required for viability, at least in yeast. By microinjection of peptides that compete and inhibit target protein phosphorylation by CKII, we demonstrate that CKII is involved in attenuating AP-1 activity *in vivo*.

Results

CKII Is a Major c-Jun Kinase

c-Jun is a nuclear phosphoprotein (Angel et al., 1989; Boyle et al., 1991a). To identify the major protein kinases that could be responsible for its phosphorylation in this compartment, we examined nuclear extracts of zinc-induced methallothionein (MT)-v-Sis/NIH 3T3 cells, NIH 3T3 subclones that are stably transformed with an MT-v-Sis fusion gene. (Essentially identical results were obtained by analyzing nuclear extracts of Ha-ras-transformed FR 3T3

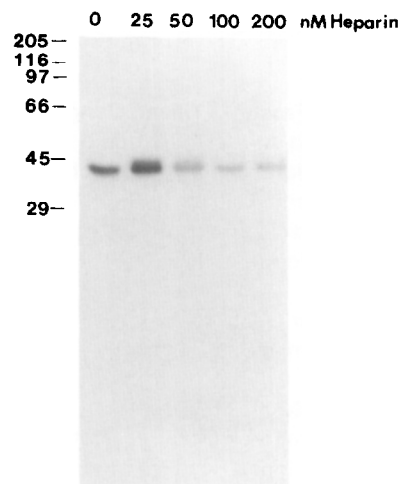


Figure 2. Inhibition of c-Jun Kinase Activity by Heparin
Recombinant c-Jun (100 ng) was incubated at 30°C for 1 hr with nuclear extract (50 μ g) of MT-v-Sis-transformed NIH 3T3 cells in the presence of 20 μ M [γ - 32 P]ATP, as described in Experimental Procedures. Increasing amounts of heparin were included as indicated. c-Jun protein was immunoprecipitated with an anti-c-Jun antiserum and resolved by SDS-PAGE, followed by autoradiography.

cells). Fractionation of these nuclear extracts on a Mono Q column revealed two major peaks of c-Jun kinase activity, determined by *in vitro* phosphorylation of a recombinant full-length c-Jun protein (Figure 1). Both peaks also contained protein kinases that phosphorylated a synthetic peptide containing 3 of the 4 potential C-terminal phosphorylation sites of c-Jun (peptide C; TPPLSPIDMESQER). Because one of the more abundant nuclear kinases is CKII and some of the C-terminal phosphorylation sites of c-Jun resemble CKII sites (Kemp and Pearson, 1990), we examined by Western blotting the presence of CKII in these fractions. Using an antibody directed against CKII holoenzyme purified from rat testes, we found that CKII antigenic activity coeluted with the major component of the second peak of c-Jun kinase activity (Figure 1C). The first peak of c-Jun kinase activity was found to correspond, at least in part, to extracellular signal-responsive kinase 1 (ERK1) and ERK2, both of which phosphorylated Ser-243 of c-Jun (A. L., M. Cobb, and M. K., unpublished data).

CKII is characterized by its very high sensitivity to inhibition by low concentrations of heparin (Hathaway et al., 1981). As shown in Figure 2, phosphorylation of c-Jun by the unfractionated nuclear extract is inhibited approximately 76% by 100 nM heparin.

CKII Is an Efficient c-Jun Kinase

The results described above suggested that CKII could be one of the major nuclear protein kinases that phosphorylate c-Jun. In support of this possibility, we found that recombinant c-Jun is phosphorylated very efficiently, with a Michaelis constant equal to 0.27×10^{-6} M, by homogeneously pure CKII (data not shown). Addition of excess peptide C competed with c-Jun phosphorylation (Figure 3). Although peptide C has migrated off the gel shown

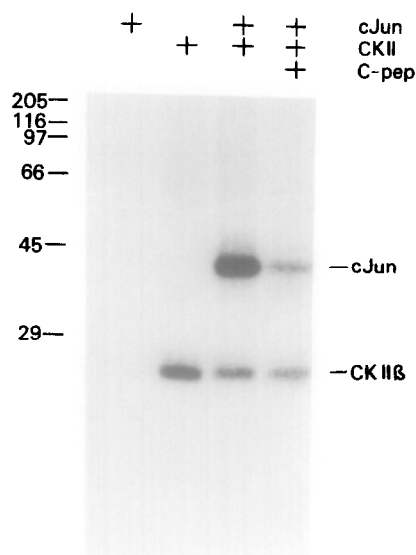


Figure 3. Phosphorylation of c-Jun by Purified CKII

Recombinant c-Jun (100 ng) was incubated in a reaction mixture at 30°C for 30 min with purified CKII (60 ng) in the presence of 20 μ M [γ - 32 P]ATP as described in Experimental Procedures. Lane 1, c-Jun alone; lane 2, CKII alone; lane 3, c-Jun and CKII; lane 4, c-Jun, CKII, and 1 mM peptide C. Proteins were resolved by SDS-PAGE and visualized by autoradiography. The bands corresponding to the c-Jun and CKII β subunit are indicated.

here, other experiments (data not shown) indicated it is efficiently phosphorylated by CKII. The stoichiometry of c-Jun phosphorylation was determined to be 1.9 mol of phosphate per mol of c-Jun protein (data not shown). In contrast to wild-type c-Jun, c-Jun(Phe-243) was phosphorylated inefficiently (approximately 10% of wild type) by CKII (Figure 4).

Phosphopeptide analysis revealed that the major phosphorylation sites of c-Jun affected by CKII are two of the C-terminal sites (Figure 5). This was indicated by the pres-

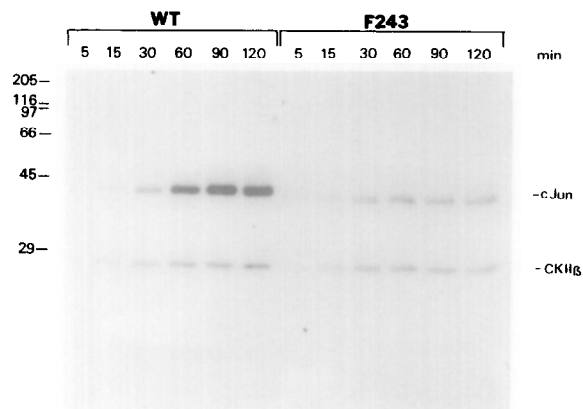


Figure 4. c-Jun(Phe-243) Is Phosphorylated by CKII Less Efficiently Than Wild-Type c-Jun

To compare the efficiency of phosphorylation by CKII of wild-type (WT) c-Jun with that of c-Jun(Phe-243), 100 ng of each protein was incubated with 150 ng of CKII in the presence of 100 μ M [γ - 32 P]ATP. At the indicated times, the reactions were stopped and the proteins were resolved by SDS-PAGE and visualized by autoradiography.

ence of spots c and b, which correspond to the tryptic peptide encompassing residues 227–252 phosphorylated on one or two sites, respectively (Boyle et al., 1991a). Together, spots b and c account for more than 90% of the 32 P incorporated into c-Jun after in vitro phosphorylation by CKII. Spot b was also a major phosphopeptide of metabolically labeled c-Jun isolated from Ha-ras-transformed FR 3T3 cells. Mixing experiments indicated that phosphopeptides b and c of in vitro phosphorylated c-Jun comigrated with phosphopeptides b and c of in vivo labeled c-Jun (data not shown). Phosphorylation of the C-terminal sites of c-Jun(Phe-243) was considerably less efficient than the phosphorylation of the same sites in wild-type c-Jun, while phosphorylation of minor sites occurred with similar efficiency on both proteins (Figure 5). Since c-Jun(Phe-243) was still phosphorylated on two of the C-ter-

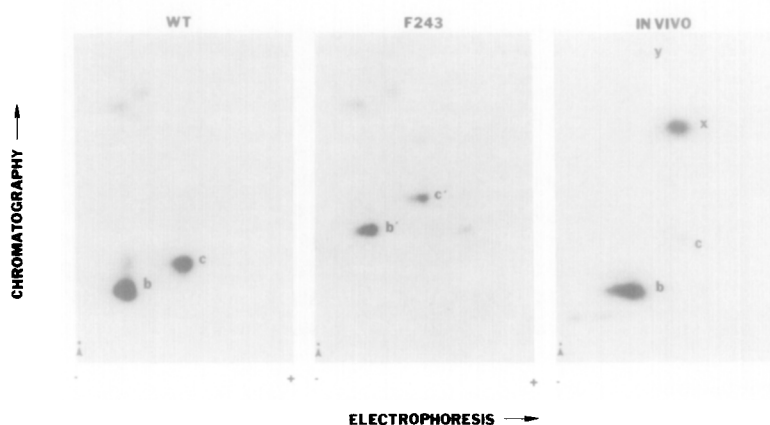


Figure 5. Two-Dimensional Tryptic Peptide Maps of In Vivo and In Vitro Phosphorylated c-Jun Proteins

Equal amounts of wild-type (WT) c-Jun and c-Jun(Phe-243) were phosphorylated in vitro with purified CKII using [γ - 32 P]ATP for 60 min. In vivo phosphorylated c-Jun protein was immunopurified from Ha-ras-transformed FR 3T3 cells labeled with [32 P]orthophosphate for 3 hr. After elution from SDS-polyacrylamide gels, the different proteins were subjected to trypsin digestion. Digests of wild-type c-Jun and c-Jun(Phe-243) phosphorylated in vitro, and in vivo labeled c-Jun were applied to thin-layer cellulose plates and resolved in the horizontal dimension by electrophoresis at pH 1.9 (anode to the left) and in the vertical dimension by ascending chromatography as described in Experimental Procedures. The plates were exposed to X-ray film for 4 days at -70°C using an intensifying screen. The origins of migration are indicated by the arrowheads.

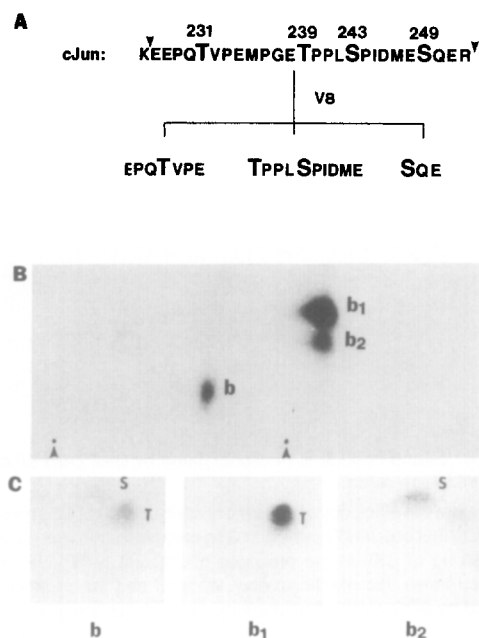


Figure 6. Determination of the Major CKII Phosphorylation Sites
(A) Shown are the sequences of the tryptic peptide containing the four potential C-terminal phosphorylation sites of c-Jun and the three phosphopeptides that can be generated by secondary digestion with V8 endopeptidase.
(B) Wild-type c-Jun (200 ng) was phosphorylated in vitro with purified CKII and subjected to tryptic peptide mapping as described in Figure 5. The phosphopeptide corresponding to spot b was extracted from the thin-layer plate and subjected to secondary digestion with V8 endopeptidase. The V8 digests were then separated in two dimensions. For comparison, a sample of the original tryptic peptide was applied to the left. The origins of migration are indicated by the arrowheads. b, tryptic peptide; b₁ and b₂, V8 digestion products; T, Thr; S, Ser. The plate was exposed at -70°C for 4 days using an intensifying screen.
(C) The b, b₁, and b₂ phosphopeptides identified in (B) were extracted from the plate and subjected to partial acid hydrolysis as described in Experimental Procedures. ³²P-labeled phosphoamino acids were detected by exposure to X-ray film and an intensifying screen for 9 days at -70°C. Nonradioactive phosphoamino acid standards were detected by ninhydrin staining.

terminal sites (as indicated by the presence of the b' and c' phosphopeptides), it appears that Ser-243 is not one of the CKII phosphorylation sites. In agreement with the higher hydrophobic moment of Phe, the b' and c' phosphopeptides derived from c-Jun(Phe-243) migrated more rapidly in the ascending direction than the b and c phosphopeptides of wild-type c-Jun (see also Boyle et al., 1991a).

To determine the exact sites phosphorylated by CKII, we used a combination of secondary digestion with V8 protease, phosphoamino acid analysis, and Edman degradation. The four potential C-terminal phosphorylation sites, Thr-231, Thr-239, Ser-243, and Ser-249, are contained within a single tryptic peptide (Figure 6A). Digestion of this peptide by V8 endopeptidase should produce three smaller phosphopeptides, one containing Thr-231 (owing to the reduced ability of V8 endopeptidase to cleave a Glu-Pro peptide bond, this peptide starts with a Glu residue), one containing Thr-239 and Ser-243, and another one containing Ser-249. V8 digestion of phosphopeptide b

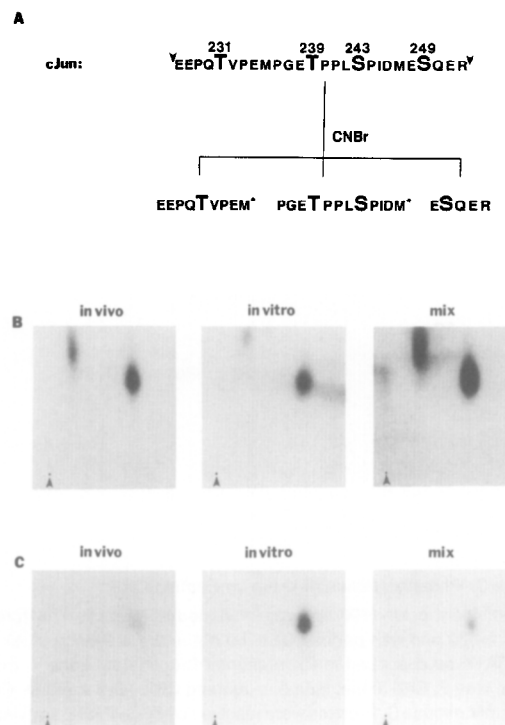


Figure 7. Comparison of In Vivo and In Vitro C-Terminal Phosphorylation Sites
(A) The sequences of the tryptic peptide containing the four potential C-terminal phosphorylation sites and the CNBr cleavage products are shown. After CNBr cleavage, the methionines (M) are converted to homo-serine lactones (M*).
(B) The b-tryptic phosphopeptides derived from ³²P-labeled c-Jun isolated from Ha-ras-transformed FR 3T3 cells (in vivo) or c-Jun phosphorylated in vitro by CKII (in vitro) was isolated after two-dimensional separation and subjected to secondary digestion with CNBr. The CNBr digestion products were subjected to another two-dimensional separation as described above.
(C) Same as described above, except that the secondary CNBr digest was performed on the c-tryptic phosphopeptides derived from in vivo and in vitro labeled c-Jun, as indicated. The plates were exposed at -70°C for 7 days using intensifying screens.

isolated from a tryptic digest of CKII-phosphorylated c-Jun produced two phosphopeptides: b₁ and b₂ (Figure 6B). Phosphoamino acid analysis indicated that b₁ contained phosphothreonine while b₂ contained phosphoserine (Figure 6C). Manual Edman degradation of phosphopeptides b₁ and b₂ resulted in the release of ³²P in the fourth and first cycles, respectively (data not shown). Therefore, the major CKII phosphorylation sites of c-Jun are Thr-231 and Ser-249. This assignment is also consistent with the relative migration positions of the b₁ and b₂ phosphopeptides. Thr-231 and Ser-249 are also residing within CKII phosphorylation consensus sequences (Pinna, 1990) TVPE and ESQE, respectively.

The CKII Sites Are Also Phosphorylated In Vivo

As mentioned above, in nonstimulated cells c-Jun is phosphorylated on three C-terminal sites that include two Ser residues and one Thr residue. By comparison with c-Jun phosphorylated in vitro by GSK3, the Ser residues were

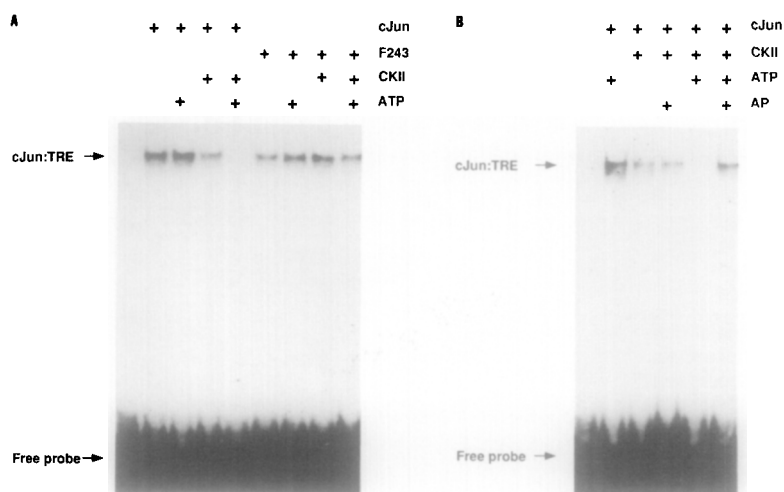


Figure 8. Differential Sensitivity of Wild-Type c-Jun and c-Jun(Phe-243) to Inhibition by CKII
Purified recombinant wild-type c-Jun or c-Jun(Phe-243) (10 ng) was preincubated at 30°C for 30 min in the absence or presence of CKII and 100 μ M ATP, as indicated. After the incubations with CKII and ATP, some protein samples were incubated with alkaline phosphatase (AP) for 30 min as indicated. The protein samples were mixed with labeled TRE probe, and DNA-protein complexes were resolved on nondenaturing polyacrylamide gels, which were dried and autoradiographed. The migration positions of the c-Jun homodimer and the labeled probe are marked by arrowheads. (A) The DNA binding of wild-type c-Jun, but not c-Jun(F243), is inhibited by CKII. (B) The inhibition of c-Jun DNA binding by CKII is reversed by alkaline phosphatase treatment.

identified as Ser-243 and Ser-249 (Boyle et al., 1991a). However, the exact Thr residue that is phosphorylated in vivo was not identified and could be either Thr-231 or Thr-239. Since in vitro Thr-239 is phosphorylated by GSK3 (Boyle et al., 1991a), whereas Thr-231 is phosphorylated by CKII, identification of the Thr residue that is phosphorylated in vivo should help in determining which of the two protein kinases is likely to phosphorylate c-Jun in vivo. To compare the sites on c-Jun that are phosphorylated in Ha-ras-transformed FR 3T3 cells with the sites phosphorylated by CKII, in vivo and in vitro phosphorylated c-Jun samples were subjected to trypsin digestion (without prior oxidation), and the b and c phosphopeptides were isolated after two-dimensional separation. The phosphopeptides were then subjected to secondary digestion with cyanogen bromide (CNBr), which should cleave the b and c phosphopeptides to yield three secondary peptides, one containing Thr-231, one Thr-239 and Ser-243, and the other Ser-249 (Figure 7A). Two-dimensional separation revealed that the CNBr cleavage products of in vivo labeled phosphopeptides b and c had identical electrophoretic and chromatographic mobilities to the CNBr cleavage products of phosphopeptides b and c derived from c-Jun phosphorylated in vitro by CKII (Figures 7B and 7C). Therefore, in rat fibroblasts c-Jun is phosphorylated on Thr-231, the same Thr residue that is phosphorylated in vitro by CKII.

Differential Regulation of c-Jun and c-Jun(Phe-243)

The results shown above indicate that only one phosphorylation site, Ser-249, is common to CKII and GSK3. Therefore, we examined whether phosphorylation by CKII can also inhibit c-Jun DNA binding. As shown in Figure 8A, phosphorylation of recombinant c-Jun by CKII resulted in a marked decrease in its DNA binding activity. By contrast, DNA binding by c-Jun(Phe-243) was only slightly inhibited by CKII, consistent with the lower efficiency of its phosphorylation. Incubation with alkaline phosphatase of c-Jun that was first phosphorylated by CKII resulted in restoration of its DNA binding activity (Figure 8B).

CKII Substrate and Inhibitory Peptides Induce AP-1 Activity

The results presented above strongly suggested that CKII is a major nuclear kinase responsible for phosphorylation of the inhibitory sites of c-Jun. To test whether CKII has a similar function in vivo, we microinjected peptides that act as competitive inhibitors of CKII into cells and examined their effect on AP-1 activity and c-jun expression. Several CKII substrates and inhibitors were synthesized. Peptide A is a highly specific CKII substrate (Kuenzel and Krebs, 1985). Peptide A1 was derived from peptide A by substituting its phosphoacceptor site by a nonphosphorylatable Ala residue. Peptide B1 was derived from a CKII substrate sequence present in avian c-Myc (Luscher et al., 1989) by substitution of its phosphoacceptor sites by Ala residues. The Ala-substituted peptides are expected to act as pseudosubstrate inhibitors of CKII (Kemp and Pearson, 1990). Peptide C, which contains two of the C-terminal phosphorylation sites of c-Jun, was described above, and peptides D and E are control peptides that correspond to a low affinity ligand of hsc70 (Flynn et al., 1989) and an antigenic epitope of polyoma middle T antigen, respectively. Peptide E contains a similar number of negative charges as peptides A, A1, and B. All peptides were first tested for their ability to inhibit c-Jun phosphorylation by CKII in vitro (Table 1). To test their effects on AP-1 activity, the peptides were microinjected into the nuclei of REF52 rat fibroblasts together with TRE-lacZ or CRE-lacZ reporter plasmids, whose expression is dependent on either AP-1- or CRE-binding protein activity, respectively, and can be determined in single cells by staining with a chromogenic β -galactosidase substrate (Meinkoth et al., 1990, 1991). Results of a typical experiment using the TRE-lacZ reporter are shown in Figure 9. We found that all four peptides that inhibited c-Jun phosphorylation by CKII in vitro (peptides A, A1, B1, and C) led to efficient induction of TRE-lacZ expression in vivo. However, peptides D and E, which did not inhibit c-Jun phosphorylation by CKII, had no effect on TRE-lacZ expression (Table 2). In addition, microinjection

Table 1. Inhibition of c-Jun Phosphorylation by Synthetic Peptides

Peptide	Sequence	Percentage of cJun Phosphorylation	
		0.1 mM	1.0 mM
A	EEETEEERRR	100	56
A1	EEEEEEERRR	60	36
B1	RRRPPAAAADAEEEQEEDDEE	100	20
C	TPPLSPIDMESQER	45	10
D	ENQFGDCHY	100	100
E	EEEEYMPME	100	100

One hundred nanograms of c-Jun was incubated with 60 ng of CKII in 35 μ l reaction mixtures as described in Experimental Procedures. Increasing concentrations of synthetic peptides were included, and their effect on c-Jun phosphorylation was determined after SDS-PAGE and autoradiography. The concentrations of the peptides are referred to as 0.1 mM and 1.0 mM.

of a protein kinase A-phosphorylated synthetic peptide that is a substrate for GSK3 but not CKII (Fiol et al., 1988) did not lead to TRE-lacZ induction (data not shown).

The induction of TRE-lacZ by the CKII substrate peptides was as efficient as the induction of the same reporter by TPA. The induction by peptide A1 was concentration dependent and was optimal at or above 300 nM, whereas induction by peptide A was optimal at or above 300 μ M. This was estimated by dilution of known concentrations of the peptide and taking into account a further 20- to 50-fold dilution within the cell (Feramisco and Welch, 1986). The difference in the potency of the two peptides is consistent with peptide A1 acting as a pseudosubstrate inhibitor and peptide A acting as a competitive substrate. Peptides A1, B1, and D did not lead to a significant induction of CRE-lacZ expression (Table 2). In addition, peptide A1 did not cause a significant inhibition of CRE-lacZ expression by cAMP. Similar results were obtained by injection of peptides A, A1, C, and D into rat-2 cells stably transfected with the reporters (data not shown). To provide further evidence that the effect of peptide A1 on TRE activity is due to inhibition of CKII, we coinjected the cells with peptide A1, the TRE-lacZ reporter, and purified CKII. These injections were intranuclear as well, and the amount of CKII that was delivered was approximately 15 fg per nucleus, which we estimate to be 30% of the total cellular content of the enzyme (approximately 50 fg per cell, as determined by Western blotting). As shown in Table 2 and in Figure 10, coinjection of CKII reversed the induction of TRE-lacZ induction by peptide A1. In addition, microinjection of the same amount of CKII inhibited the induction of TRE-lacZ by TPA. CKII microinjection, however, did not have a significant effect on the induction of CRE-lacZ expression by cAMP.

Injection of peptide A1, but not D, also led to induction of c-Jun, but not c-Fos, expression, as determined by indirect immunofluorescence using anti-c-Jun and anti-c-Fos antibodies (Figure 11). Previous results indicate that *c-jun* expression is positively autoregulated by AP-1 (Angel et al., 1988b). Therefore, all of these results strongly suggest that inhibition or titration of CKII in vivo leads to induction of AP-1 activity.

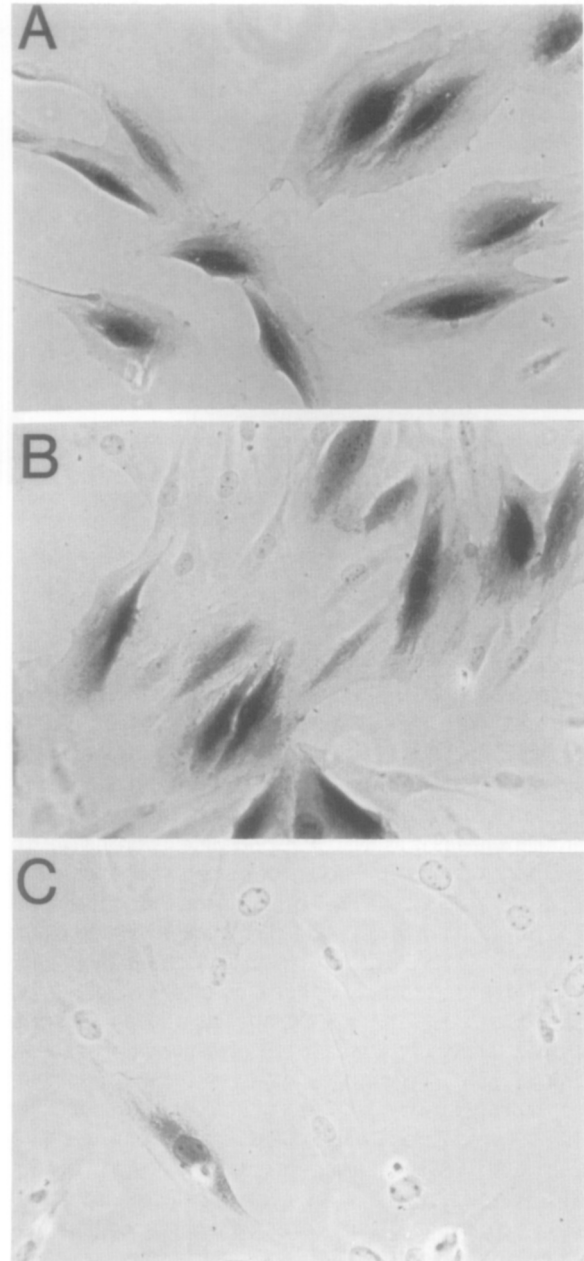


Figure 9. Microinjection of CKII Substrate Peptides Induces AP-1 Activity

Nuclei of quiescent REF52 cells were microinjected with the TRE-lacZ reporter plasmid and synthetic CKII inhibitor A1 peptide (A), TRE-lacZ- and c-Jun-derived C peptide (B), or TRE-lacZ and low affinity hsp70 ligand D peptide (C). Rat IgG was coinjected into all cells to allow the identification of injected cells by indirect immunofluorescence (not shown). Virtually all of the cells in the fields shown were microinjected. The stained cells are positive for β -galactosidase activity.

Discussion

AP-1 activity is subject to complex regulation involving both posttranslational modulation of Jun and Fos protein activity and transcriptional regulation of the *jun* and *fos* genes (reviewed in Karin, 1991; Angel and Karin, 1991). In nonstimulated fibroblasts or epithelial cells, the level of

Table 2. Effects of Microinjected Peptides on Reporter Gene Expression

Reporter	Treatment	Percentage of Blue Cells
TRE	A	69 ± 3
TRE	A1	84 ± 4
TRE	B1	65 ± 4
TRE	C	62 ± 11
TRE	D	28 ± 10
TRE	E	16 ± 2
TRE	TPA	76 ± 10
TRE	IgG	23 ± 8
TRE	A1 + CKII	18 ± 9
TRE	TPA + CKII	26 ± 15
TRE	IgG + CKII	21 ± 6
CRE	A	—
CRE	A1	13.5
CRE	B1	16.5
CRE	C	—
CRE	D	14.6
CRE	cAMP	59.6
CRE	cAMP ± A1	53.0
CRE	IgG	2.7
CRE	cAMP + CKII	51.8

In a typical experiment, at least 100 cells were injected intranuclearly. All peptides were injected at 5 mg/ml with the exception of peptide C, which was injected at 4 mg/ml. Where indicated, purified CKII (0.4 mg/ml) was also included in the injection and the A1 peptide was preincubated with the enzyme for 45–60 min prior to injection. Each cell received approximately 15 fg of CKII. All of the TRE–lacZ values are the averages of three separate experiments, with the exception of peptides A and E and the TPA stimulation, each of which was done twice. The CRE–lacZ values are the averages of two separate experiments. Cells were stimulated with TPA at a final concentration of 100 ng/ml. For cAMP stimulation, 1 nM isobutylmethylxanthine and 1 nM 8-Br-cAMP were used.

Fos protein expression is extremely low, and therefore the majority of AP-1 activity is likely to be composed of Jun homo- and heterodimers (Chiu et al., 1988; Kovary and Bravo, 1991; Alani et al., 1991). Among the Jun proteins, c-Jun is the most effective transcriptional activator (Yang-Yen et al., 1990), and despite its inducible expression (Angel et al., 1988b), all fibroblastic and epithelial cells analyzed contain basal levels of c-Jun (Angel et al., 1988b; Boyle et al., 1991a). In such cells, c-Jun is phosphorylated on two to three sites, located immediately upstream of its DNA-binding domain. Phosphorylation of these C-terminal sites inhibits c-Jun DNA binding (Boyle et al., 1991a). The experiments described above identify a major protein kinase responsible for phosphorylation of the inhibitory C-terminal sites as CKII. This conclusion rests on four lines of evidence.

First, fractionation of nuclear extracts suggests that CKII is the major nuclear kinase that phosphorylates the C-terminal sites of c-Jun. The phosphorylation of c-Jun by unfractionated nuclear extracts is markedly inhibited by low concentrations of heparin, which is consistent with the known high sensitivity of CKII to inhibition by heparin (Hathaway et al., 1981).

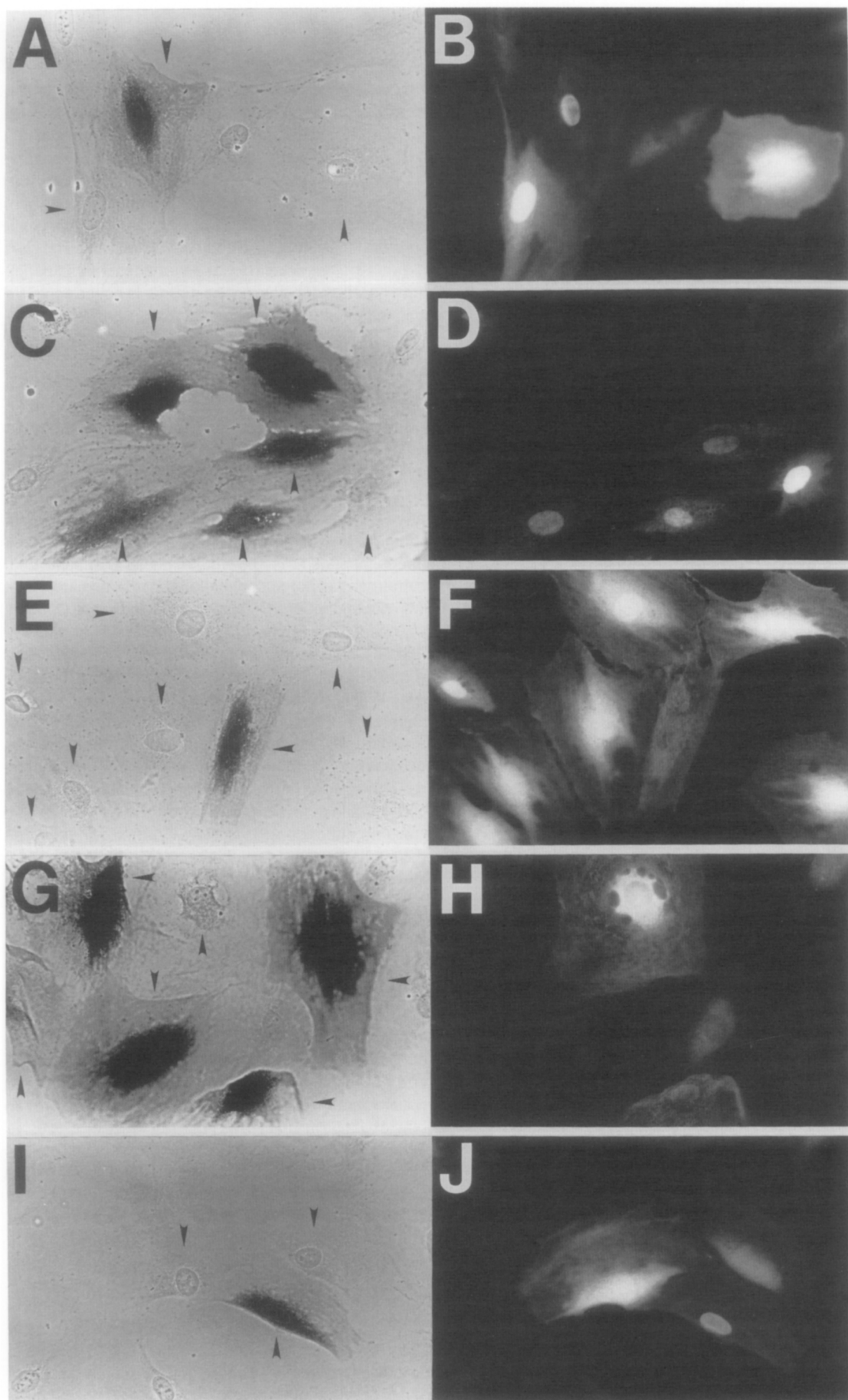
Second, while CKII phosphorylates wild-type c-Jun very efficiently, it phosphorylates inefficiently a mutant c-Jun containing Phe instead of Ser at position 243. Whereas

DNA binding by wild-type c-Jun is inhibited by CKII, binding by c-Jun(Phe-243) is unaffected. In vivo, this amino acid substitution decreases phosphorylation of the remaining C-terminal sites (Boyle et al., 1991a; Smeal et al., 1991). Since the two phosphoacceptor sites phosphorylated by CKII, Thr-231 and Ser-249, are still present in c-Jun(Phe-243), these findings indicate that CKII is as sensitive to substitution of Ser-243 by Phe as is the physiological protein kinase that phosphorylates the C-terminal sites of c-Jun. Another protein kinase that phosphorylates the C-terminal sites of c-Jun in vitro, GSK3, is not as sensitive to this mutation (Boyle et al., 1991a).

Third, in addition to Ser-243 and Ser-249, the in vivo C-terminal phosphorylation sites of c-Jun include Thr-231. The same Thr residue is phosphorylated in vitro by CKII but not by GSK3.

Fourth, microinjection of peptides that act as inhibitors of c-Jun phosphorylation by CKII in vitro into serum-starved rat fibroblasts results in a marked increase in AP-1 activity measured by a coinjected reporter plasmid. These peptides are as efficient in inducing the AP-1-responsive reporter as is treatment with TPA. At least one of these peptides, peptide A, is known to be a highly specific CKII substrate and is not phosphorylated by protein kinase A, cGMP-dependent protein kinase, protein kinase C, phosphorylase kinase, myosin light chain kinase, and microtubule-associated protein kinase/ERK (Kuenzel and Krebs, 1985). Therefore the effect of peptide A on AP-1 activity is likely to be due to its competition with nuclear substrates for phosphorylation by CKII and not by other protein kinases. By conjecture, peptide A1 that is a non-phosphorylatable derivative of peptide A should function as a highly specific, competitive pseudosubstrate inhibitor of CKII. Peptides that are not recognized by CKII, D and E, have no effect on AP-1 activity. Further evidence that peptide A1 acts in vivo by inhibition of CKII is the reversal of its effect on AP-1 activity by coinjection of purified CKII. In addition, microinjection of peptide A1, but not D, leads to induction of c-Jun expression. Neither peptide has a measurable effect on c-Fos expression. These results are consistent with the existing model of positive autoregulation of c-Jun expression by AP-1 (Angel and Karin, 1991) and provide evidence that the peptide inhibitors also lead to induction of endogenous AP-1-dependent genes. However, it remains to be demonstrated that these peptides interfere with c-Jun phosphorylation in vivo. Our initial attempts to demonstrate an effect of the peptides on protein phosphorylation in intact cells have not yielded conclusive results, owing to technical difficulties associated with the analysis of specific protein phosphorylation in a small number of cells (100 cells per experiment).

Collectively, the results presented in this paper suggest that in resting fibroblasts, AP-1 activity is attenuated by phosphorylation of c-Jun by CKII on two inhibitory C-terminal sites (Thr-231 and Ser-249). In response to TPA stimulation, at least one or both of these sites are dephosphorylated (Boyle et al., 1991a) and thereby increase c-Jun binding activity. As shall be discussed below, CKII is a constitutive enzyme, and therefore TPA stimulation appears to result in activation of an as yet to be identified



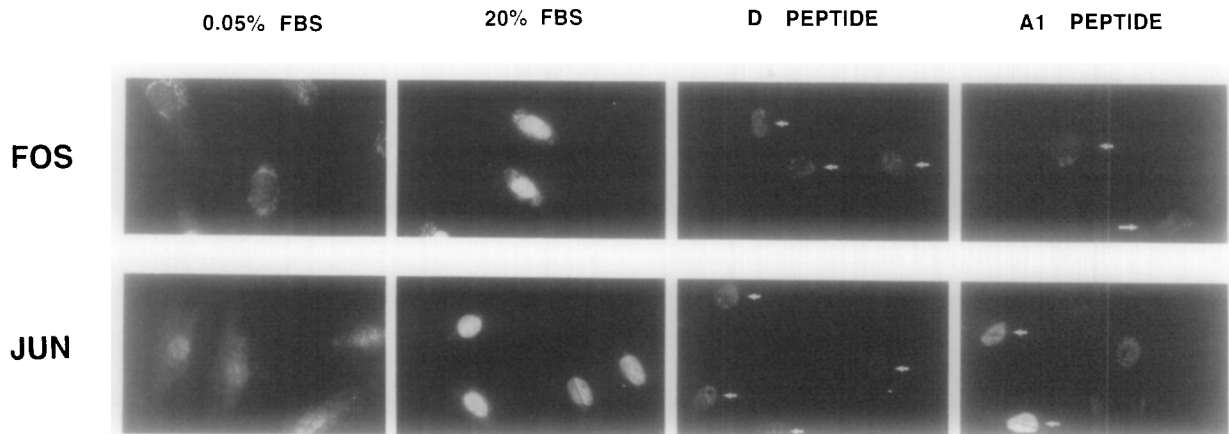


Figure 11. Microinjection of CKII Pseudosubstrate Stimulates Expression of c-Jun but Not c-Fos

Quiescent REF52 cells were microinjected with either the A1 or D peptides, and endogenous c-Jun or c-Fos expression was determined by indirect immunofluorescence, using anti-c-Jun and anti-c-Fos antibodies, 120 min and 75 min later, respectively. Injected cells are marked by arrows. The peptides were injected at either 20 mg/ml or 5 mg/ml when assaying for c-Fos or c-Jun expression, respectively. The two left lanes show endogenous c-Jun and c-Fos expression in quiescent and fetal bovine serum (FBS)-stimulated REF52 cells.

phosphatase. Since both CKII sites are fully conserved in JunD, it is possible that JunD is subject to similar regulation. Because CKII does not phosphorylate c-Jun on Ser-243, one of the *in vivo* C-terminal phosphorylation sites (Boyle et al., 1991a), it appears that this site is phosphorylated by another protein kinase that remains to be identified. However, the phosphorylation of Ser-243 in Ha-ras-transformed FR 3T3 cells appears to be low and therefore of questionable function. One possible kinase-phosphorylating Ser-243 could be ERT, which was shown to phosphorylate Ser-243 *in vitro* (Alvarez et al., 1991) or the related ERK1 and ERK2 (A. L. and M. Cobb, unpublished data; J. Ferrell, personal communication). Although it is possible that phosphorylation of Ser-243 by ERK may potentiate the phosphorylation of the two flanking sites by CKII, so far we found no effect of prephosphorylation with ERK1 and 2 on the kinetics of c-Jun phosphorylation by CKII (A. L., unpublished data). However, recent work has shown that a phosphoserine residue that follows a phosphorylatable amino acid can serve as a specificity determinant for CKII (Litchfield et al., 1990a), and therefore it is still possible that the different phosphorylation sites on the C-terminus of c-Jun may affect each other. It is of interest that Ser-243 is converted to Phe in v-Jun (Maki et al., 1987; Angel et al., 1988a). The same substitution, when introduced into c-Jun, decreases phosphorylation of all C-terminal sites while increasing its *in vivo* activity (Boyle et al., 1991a). It is plausible that Phe-243 interferes with phosphorylation of the two flanking sites by CKII, either by steric hindrance or by altering the conformation of the

protein in this region and decreasing its accessibility to the protein kinase. An additional effect of Phe-243 could be increased sensitivity to phosphatase action.

c-Jun is not the only DNA-binding protein that is inhibited by CKII. Another such protein is c-Myb, which is phosphorylated *in vitro* on two sites, Ser-11 and Ser-12, by CKII. Phosphorylation of these sites decreases the DNA binding activity of c-Myb but has no effect on v-Myb, which lost both of these residues (Luscher et al., 1990). Although Ser-11 and Ser-12 are also phosphorylated *in vivo*, there is no proof yet that CKII is responsible for their phosphorylation or that phosphorylation of these sites controls c-Myb activity *in vivo*.

Another potential substrate for phosphorylation by CKII is SRF (Manak et al., 1990; Gauthier-Rouvière et al., 1991). Phosphorylation of SRF by CKII increases its ability to bind to the c-fos serum response element *in vitro* (Manak et al., 1990), while microinjection of CKII can lead to c-fos induction *in vivo* (Gauthier-Rouvière et al., 1991). Other investigators, however, found only a moderate effect of phosphorylation by CKII on SRF binding activity (Marais et al., 1992). Furthermore, genomic footprinting experiments indicate that occupancy of the c-fos serum response element does not change before or after induction by serum (Herrera et al., 1989). In addition, the state of SRF phosphorylation is not affected by serum stimulation (R. Prywes, personal communication). If CKII is indeed required for c-fos induction, our findings indicate that it is possible to induce AP-1 activity without inducing expression of c-Fos because competition for or inhibition of CKII

Figure 10. Inhibition of TRE-lacZ Expression by Nuclear Injection of CKII

Nuclei of quiescent REF52 cells were microinjected with the TRE-lacZ reporter plasmid and rabbit IgG (A and B). In addition, cells were treated with TPA (100 ng/ml) for 2 hr (C and D), TPA + CKII microinjection (E and F), microinjection of peptide A1 (G and H), or microinjection peptide of A1 + CKII (I and J). (A), (C), (E), (G), and (I) show the results of the β -galactosidase staining, while (B), (D), (F), (H), and (J) show the results of indirect immunofluorescence with anti-rabbit IgG antibody. Note that high levels of β -galactosidase staining quench the fluorescent signal. All of the microinjected cells are indicated by the arrowheads.

activity is expected to prevent c-Fos synthesis. Furthermore, we found that intranuclear injection of CKII led to inhibition rather than induction of AP-1 activity. In addition, microinjection of peptide A1, the most potent inducer of AP-1 activity, did not lead to c-Fos induction. Other examples that AP-1 activity can be stimulated in the absence of c-Fos induction are provided by experiments in which c-Fos synthesis was blocked by protein synthesis inhibitors (Devary et al., 1991) or by selective protein kinase inhibitors such as 2-aminopurine (Mahadevan et al., 1990).

The conclusion that CKII attenuates AP-1 activity appears to be contradictory to earlier reports that CKII activity is elevated after cell stimulation with growth factors (Sommercorn et al., 1987; Klarlund and Czech, 1988; Ackerman and Osheroff, 1989; Carroll et al., 1988; Ackerman et al., 1990). However, recent studies have failed to detect measurable effects of growth factors on CKII activity (E. Krebs, personal communication; M. Czech, personal communication), phosphorylation (U. K. and T. H., unpublished data), or subcellular localization (Krek et al., 1992). In fact, it has been suggested that CKII is responsible for constitutive phosphorylation of a number of important substrates and that regulation is achieved by protein phosphatases 1 and 2A, whose activity is modulated by extracellular signals (Pinna, 1990). This suggestion appears to apply for c-Jun: in resting cells c-Jun is constitutively phosphorylated by CKII, and upon TPA treatment or expression of a variety of transforming oncoproteins, it is rapidly dephosphorylated on at least one of the inhibitory sites next to its C-terminal DNA-binding domain (Boyle et al., 1991a; Binetruy et al., 1991; Smeal et al., 1991). Dephosphorylation increases c-Jun DNA binding activity, resulting in elevated transcription of c-Jun-responsive genes. Since one such gene is *c-jun* itself (Angel et al., 1988b), the initial dephosphorylation event can result in a further and more sustained increase in AP-1 activity. In agreement with this model, we find that microinjection of the pseudosubstrate inhibitor of CKII, peptide A1, leads to induction of c-Jun expression. In addition, intranuclear injection of CKII inhibits the induction of AP-1 activity by TPA.

Although CKII was originally thought to reside in several cellular compartments (reviewed in Meisner and Czech, 1991), recent evidence indicates that it is mostly nuclear (Krek et al., 1992). This suggests that the majority of its substrates are nuclear proteins. Indeed, CKII has been shown to phosphorylate a large number of nuclear proteins in vitro (reviewed in Meisner and Czech, 1991). However, its physiological substrates and functions remained unknown. Yet there is no doubt that CKII is a very important enzyme, as disruption of the α and α' subunit genes in yeast results in lethality (Padmanabha et al., 1990). This absolute requirement of CKII activity for cell viability had, however, impeded the use of genetic analysis to identify its biological role. The work described here establishes one role for CKII, which is the attenuation of AP-1 activity in nonstimulated fibroblasts. The role of CKII in controlling the activity of other nuclear factors remains to be determined. However, the approach described here based on titration or inhibition of CKII activity by microinjected substrate and inhibitor peptides should be of general utility.

Experimental Procedures

Materials

Carrier-free [32 P]orthophosphate was obtained from ICN Pharmaceuticals, while [γ - 32 P]ATP (3000 mCi/nmol) was from Amersham Corporation. TPCK-trypsin (Worthington Biochemical Corporation) was stored at 1 mg/ml. Staphylococcus aureus V8 endopeptidase (ICN Immunochemicals) was stored at 5 mg/ml. Calf intestinal alkaline phosphatase (Boehringer Mannheim Biochemicals) was stored at 1 U/ μ l. CKII (a gift from D. Litchfield and E. Krebs) was purified from bovine testes. All enzymes were stored at -80°C . The following peptides were used: specific CKII substrate and its derived inhibitor (EEETEEERRR and EEEAEEERRR) (Kuenzel and Krebs, 1985); c-Myc-derived CKII inhibitor (RRPPAAAADAEQEEDEE) (Luscher et al., 1989); c-Jun C-terminal peptide (TPPLSPIDMESQER); hsc70 low affinity ligand (ENQFGDCHY) (Flynn et al., 1989); an antigenic epitope of polyoma middle T antigen (EEEEYMPME) (Tooze, 1980). All peptides were synthesized on an Applied Biosystems peptide synthesizer using fluorenylmethoxycarbonyl chemistry and were purified by high pressure liquid chromatography on a C18 column, except for the hsc70 low affinity peptide, which was purified on a G10 gel filtration column.

Cell Culture, Labeling, and Immunoprecipitation

Ha-ras-transformed FR 3T3 cells were grown in Dulbecco's modified Eagle's medium supplemented with 10% fetal calf serum. MT-v-Sis-transformed NIH 3T3 cells (Smeal et al., 1992) were grown in Dulbecco's modified Eagle's medium supplemented with 5% newborn calf serum in the presence of 50 μM zinc acetate. Subconfluent cultures of Ha-ras-transformed FR 3T3 cells were metabolically labeled with [32 P]orthophosphate as described (Smeal et al., 1991). Labeled cells were lysed in RIPA buffer (Hunter and Sefton, 1980), and c-Jun protein was immunoprecipitated from the lysates with rabbit antiserum directed against a C-terminal peptide of human c-Jun (Binetruy et al., 1991). Immunoprecipitates were washed three times with RIPA buffer, resuspended in 1 $\times$ Laemmli sample buffer (Laemmli, 1970), boiled for 4 min, and resolved on 10% SDS-polyacrylamide gels. The gels were dried and exposed to Kodak X-OMAT-AR film, the visualized c-Jun bands were excised, and the ^{32}P -labeled protein was eluted for peptide mapping.

Preparation and Fractionation of Nuclear Extracts

Subconfluent cultures of MT-v-Sis-transformed NIH 3T3 cells were harvested and nuclear extracts were prepared as described (Bodner and Karin, 1987) in the presence of protease (10 $\mu\text{g}/\text{ml}$ leupeptin, 10 $\mu\text{g}/\text{ml}$ aprotinin, 10 $\mu\text{g}/\text{ml}$ pepstatin, 0.1 mM phenylmethylsulfonyl fluoride) and phosphatase (20 mM β -glycerophosphate, 20 mM p-nitrophosphate, 50 μM Na_2VO_4) inhibitors. Five milligrams of nuclear extract protein was chromatographed on a Mono Q anion exchange high pressure liquid chromatography column (HR5/5) that was pre-equilibrated with 50 mM Tris-HCl (pH 7.5), 20 mM β -glycerophosphate, 5 mM MgCl_2 , 10 mM β -mercaptoethanol, 1 mM EDTA, 1 mM EGTA (buffer A). Proteins were eluted with a linear gradient of 0.01–0.6 M NaCl in buffer A. Fractions of 0.5 ml were collected and assayed for c-Jun kinase activity using purified full-length recombinant c-Jun protein or peptide C as substrates.

Samples of the same fractions were separated on 10% SDS-polyacrylamide gel electrophoresis (SDS-PAGE), transferred to nitrocellulose, probed with a polyclonal antibody raised against CKII purified from rat testes (U. K. and T. H., unpublished data), and visualized using anti-rabbit antibody conjugated to horseradish peroxidase as described by the supplier (Amersham).

Phosphorylation Assays

Phosphorylation of c-Jun proteins by CKII was done at 30°C in a reaction mixture (35 μl) containing 50 mM HEPES (pH 7.6), 150 mM NaCl, 10 mM MgCl_2 . Phosphorylation was initiated by adding 5 μl of 350 μM [γ - 32 P]ATP (4,000–10,000 cpm/pmol). The reaction was terminated with the addition of 11.7 μl of 4 $\times$ Laemmli sample buffer and boiled for 4 min. Proteins were resolved on 10% SDS-polyacrylamide gel followed by autoradiography. Quantitation of phosphate incorporation was determined by counting the radioactivity of gel slices.

Phosphorylation of c-Jun protein by nuclear extract fractions was done at 30°C in a reaction mixture (35 μl) containing 50 mM HEPES

(pH 7.6), 20 mM β -glycerophosphate, 2 mM dithiothreitol, and an appropriate amount of protein in the presence of 50 μ M [γ - 32 P]ATP (4,000–10,000 cpm/pmol). The reaction was terminated by adding 2% SDS and 10% Nonidet P-40 (NP-40) (final concentrations = 0.1% and 0.5%, respectively). The c-Jun protein was immunoprecipitated and analyzed by 10% SDS-PAGE followed by autoradiography. When using peptide C as a substrate, the phosphorylation reaction was terminated by spotting 25 μ l of the reaction mixture on a Whatman P81 filter paper that was immediately immersed in 0.5% phosphoric acid as described (Cicirelli et al., 1988). The P81 paper was washed three times in 0.5% phosphoric acid, 15 min per wash, air dried, and counted in a liquid scintillation counter. Control reactions were performed under identical conditions without the peptide and used as background. The expression, purification, and renaturation of the full-length c-Jun and c-Jun(Phe-243) (Boyle et al., 1991a) are described in detail elsewhere (Deng and Karin, 1992).

Mobility Shift Assay

Recombinant c-Jun or c-Jun(Phe-243) DNA binding activity was measured by mobility shift assay as described (Smeal et al., 1989). In brief, protein-DNA complexes were formed at 23°C for 20 min in 20 μ l of reaction buffer containing 10 mM HEPES (pH 7.6), 50 mM KCl, 0.1 mM EDTA, 5 mM MgCl₂, 5 mM dithiothreitol, 10% glycerol, 1 mg/ml bovine serum albumin, 100 μ g/ml poly(dI-dC), and 0.1 ng of 32 P-labeled and filled-in TRE-containing DNA (1.0 $\times$ 10⁹ cpm/ μ g). The target DNA was the double-stranded TRE-containing oligonucleotide:

AGCTTGGTGACTCATCG
ACCACTGAGTAGGCTAG

Specificity of binding was determined by competition experiments using excess of the unlabeled oligonucleotide described above. Protein-DNA complexes were resolved on 5% nondenaturing polyacrylamide gel containing 0.4 $\times$ TBE buffer (pH 7.9) at 200–250 V for 2.5 hr at 4°C. At least 90% of the protein was active in DNA binding. Prior to the mobility shift assay, c-Jun proteins (10 ng) were modified by phosphorylation in 10 μ l of CKII reaction mix, as described above, at 30°C for 30 min, and then chilled on ice. For dephosphorylation experiments, 0.5 U of calf intestinal alkaline phosphatase was added to the reaction mixture after the incubation with CKII, further incubated at 30°C for 30 min, and then chilled on ice.

Phosphopeptide Mapping and Phosphoamino Acid Analysis

c-Jun protein phosphorylated *in vivo* or *in vitro* was purified from 10% SDS-polyacrylamide gels visualized by autoradiography. Eluted c-Jun protein was subjected to trypsin digestion as described (Hunter and Sefton, 1980). Tryptic digests were applied to thin-layer cellulose plates for two-dimensional peptide mapping by electrophoresis at pH 1.9 for 45 min at 1.0 kV in the first dimension and then by ascending chromatography in the second dimension (Boyle et al., 1991b). The phosphoamino acid composition of c-Jun phosphopeptide b was determined by two-dimensional electrophoresis of a partial acid hydrolysate (Hunter and Sefton, 1980). The position of phosphoamino acid residues in V8 subfragments of phosphopeptide b was identified by manual Edman degradation as previously described (Hunter et al., 1984). Secondary digestion by CNBr was done by incubation of tryptic peptides derived from nonoxidized c-Jun with 300 mg/ml CNBr in 70% formic acid at room temperature for 24 hr in the dark.

Microinjection and Analysis of c-Jun and c-Fos Expression

REF52 cells were grown on glass coverslips in Dulbecco's modified Eagle's medium supplemented with 10% fetal bovine serum to approximately 40% confluency, at which time they were serum starved for 24–36 hr in Dulbecco's modified Eagle's medium plus 0.05% fetal calf serum. In a typical experiment, approximately 100 cells were injected with each peptide. All peptides, except the c-Jun-derived peptide C, were injected at 5 mg/ml in 100 mM KCl, 5 mM Na₂HPO₄ (pH 7.2), unless otherwise stated. Peptide C was injected at 4 mg/ml. TRE or CRE reporter plasmids were coinjected at 0.5 mg/ml, and 5–10 mg/ml rat immunoglobulin G (IgG) was included as a marker for injected cells. The TRE-lacZ and CRE-lacZ reporters consist of four tandem repeats of either the collagenase TRE or human vasoactive intestinal polypeptide CRE sequences cloned upstream of a truncated Rous sarcoma

virus promoter and the *Escherichia coli lacZ* gene. This Rous sarcoma virus promoter fragment is inactive without the adjacent enhancer elements (Meinkoth et al., 1990). The cells were fixed 2 hr after injection for 5 min with 3.7% formaldehyde in phosphate-buffered saline (PBS), and then washed extensively with PBS. β -Galactosidase activity was detected by staining overnight with X-Gal (Kodak). The following day, the cells were permeabilized with 0.3% Triton X-100 in PBS, washed, and stained for injected rat IgG with rhodamine-conjugated donkey anti-rat IgG antibody (Jackson Laboratories) diluted 1:100 in PBS plus 0.5% NP-40 and a 2 mg/ml concentration of either bovine serum albumin or REF52 whole-cell extract. Phase contrast and fluorescence microscopy were performed using a Zeiss Axiophot microscope.

Endogenous c-Jun and c-Fos expression was determined by indirect immunofluorescence using affinity-purified rabbit polyclonal antibodies specific for c-Fos and c-Jun (Oncogene Science). The levels of c-Fos and c-Jun expression were determined in cells that were fixed 75 min or 120 min after injection, respectively. When staining for c-Fos, cells were permeabilized with 0.3% Triton X-100 in PBS for 5 min, followed by incubation with 10% goat serum in 0.5% NP-40 and PBS for 10 min. Anti-c-Fos antibody was then used at a 1:100 dilution in REF52 whole-cell extract plus 0.5% NP-40 and 1 mg/ml bovine serum albumin, followed by staining with fluorescein isothiocyanate- or rhodamine-conjugated goat anti-rabbit antibody (Jackson Laboratories). c-Jun staining was performed as described for c-Fos, except that the anti-c-Jun antibody was diluted 1:25 in 10% goat serum plus 0.5% NP-40.

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